■ Mushroom Business Management & E-Commerce Platform (Mushroom ERP)

Document 07: Business Domain Logic & Operational Workflow Specifications

Project Name: SporeERP
Document Code: DOC-07-BIZ-LOGIC
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Domain Focus: Agronomy State Machine, Double-Entry Inventory, Order Lifecycle & P&L Engine

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1. End-to-End Mushroom Agronomy & Cultivation Lifecycle

Mushroom cultivation requires precise biological stage transitions. The `ProductionBatch` state machine governs each phase:

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2. Atomic Customer Checkout & Concurrency Locking Flow

To guarantee that fresh perishable mushrooms are never oversold during high-traffic flash sales or morning harvest drops, checkout executes as an atomic database transaction:

3. Wastage, Spoilage & Biological Contamination Handling

3.1 Farm-Floor Contamination (Beds)

When agronomy staff discovers green mold (*Trichoderma*), ink cap (*Coprinus*), or bacterial blotch during daily bed inspection:
1. Staff marks bed status as `CONTAMINATED` in the Growing Room Bed Grid (`/admin/rooms`).
2. Bed is immediately quarantined and removed from the room to prevent airborne spore cross-contamination.
3. System logs `CONTAMINATION_LOSS` in `ProductionBatch` :
$$\text{Batch Contamination Rate (\%)} = \left(\frac{\text{Contaminated Beds}}{\text{Total Initial Beds}} \right) \times 100$$
4. Cost of lost spawn and substrate is automatically reflected in the batch cost accounting.

3.2 Post-Harvest Fresh Spoilage (Cold-Chain Expiry)

Fresh mushrooms unsold within **48 hours** are automatically flagged by the FIFO freshness monitor:
* **Option A (Dehydration Conversion):** Staff routes unsold Grade A mushrooms into the dehydrator. System debits Fresh Inventory and credits Dry Mushroom Inventory at a **10:1 ratio** (10kg fresh = 1kg dry mushroom).
* **Option B (Wastage Write-Off):** If spoiled, staff records `WASTAGE_SPOILAGE` in `inventory_transactions` . System logs exact financial write-off value (`Quantity * Cost per KG`).

4. Multi-Tier B2B Pricing & Bulk Quotation Engine

The catalog dynamically renders prices based on customer account classification:

- **Credit Terms for B2B:** Approved restaurants can place orders on a `NET_15` or `NET_30` invoice cycle up to their assigned `credit_limit` . Orders automatically generate GST-compliant tax invoices with HSN code `07095900` .

5. Real-Time Profit & Loss (P&L) Calculation Engine

The financial engine aggregates operational data dynamically without batch delays: